

NP4: Knapsack

Knapsack Lecture Outline

1. NP-completeness:
 - All of NP $\rightarrow$ SAT $\rightarrow$ 3SAT $\rightarrow$ IS
 - 3SAT $\rightarrow$ Subset-Sum $\rightarrow$ Knapsack
 - IS $\rightarrow$ Clique, Vertex Cover
2. DP solution to knapsack is $O(nB)$, not polynomial

Subset Sum

1. Input: Positive integers $a_1, \dots, a_n$ and t
2. Output: Subset S of $\{1, \dots, n\}$ where:
 - $\text{sum}(a_i) = t$
 - Return NO if no such subset exists
3. Dynamic programming: Give an $O(nT)$ algorithm

Subset Sum in P?

1. Claim: Subset-sum is known to be in P
 - False; subset-sum is not known to be polynomial in the input size

Subset-Sum NP-complete

1. Theorem: Subset-sum problem is NP-complete
 - Subset-sum is in NP
 - Given inputs $a_1, \dots, a_n$, t and S , check that $\text{sum}(a_i) = t$, which takes $O(n \log t)$
 - Reduce 3SAT to subset-sum

3SAT Subset-Sum: Input

1. Input to subset-sum: $2n + 2m + 1$ numbers
 - $v_1, v_1', \dots, v_n, v_n'$ and t
 - All are $\leq n + m$ digits long and base 10
 - t is on the order of 10^{n+m}

3SAT Subset-Sum: Variables

1. v_i corresponds to x_i : $v_i \text{ in } S \iff x_i = T$
2. v_i' corresponds to $\neg x_i$: $v_i' \text{ in } S \iff x_i = F$
 - Need to ensure exactly one of v_i or v_i' is in S
 - In i^{th} digit of v_i, v_i' and t put a 1
 - All other numbers put a 0

3SAT Subset-Sum: Example

Example reduction

$$f = (\overline{x_1} \vee \overline{x_2} \vee \overline{x_3}) \wedge (\overline{x_1} \vee \overline{x_2} \vee x_3) \wedge (x_1 \vee \overline{x_2} \vee x_3) \wedge (x_1 \vee x_2)$$

Input
to
subset-sum

	x_1	x_2	x_3	C_1	C_2	C_3	C_4
v_1	1	0	0				
v_1'	1	0	0				
v_2	0	1	0				
v_2'	0	1	0				
v_3	0	0	1				
v_3'	0	0	1				
S_1	0	0	0				
S_1'	0	0	0				
S_2	0	0	0				
S_2'	0	0	0				
S_3	0	0	0				
S_3'	0	0	0				
S_4	0	0	0				
S_4'	0	0	0				
T	1	1	1				

Example

3SAT Subset-Sum: Clauses

1. Digit $n+j$ corresponds to clause C_j
 - If x_i in C_j , put a 1 in digit $n+j$ for v_i
 - If $\neg x_i$ in C_j , put a 1 in digit $n+j$ for v_i'
2. Use S buffer numbers to get to the desired sum for some digit

Example reduction

$$f = (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_3) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee x_3) \wedge (x_1 \vee \bar{x}_2 \vee x_3) \wedge (x_1 \vee x_2)$$

Input
to
subset-sum

	x_1	x_2	x_3	C_1	C_2	C_3	C_4	
v_1	1	0	0	0	0	1	1	$v_1 = 1000011$
v_1'	1	0	0	1	1	0	0	$v_1' = 1001100$
v_2	0	1	0	0	0	0	1	
v_2'	0	1	0	1	1	1	0	
v_3	0	0	1	0	1	1	0	
v_3'	0	0	1	1	0	0	0	
S_1	0	0	0	1	0	0	0	
S_1'	0	0	0	1	0	0	0	
S_2	0	0	0	0	1	0	0	
S_2'	0	0	0	0	1	0	0	
S_3	0	0	0	0	0	1	0	
S_3'	0	0	0	0	0	1	0	
S_4	0	0	0	0	0	0	1	$S_4 = 1$
S_4'	0	0	0	0	0	0	1	$S_4' = 1$
Σ	1	1	1	3	3	3	3	$\Sigma = 1113333$

Example

3SAT Subset-Sum: Buffers

- Digit $n+j$ corresponds to clause C_j
 - If x_i in C_j , put a 1 in digit $n+j$ for v_i
 - If $\neg x_i$ in C_j , put a 1 in digit $n+j$ for v_i'
- Put a 3 in digit $n+j$ of t
 - Use S_j, S_j' as buffers:
 - Put a 1 in digit $n+j$ of S_j and S_j'
 - Put a 0 in digit $n+j$ of other numbers

3SAT Subset-Sum: Correctness

- Subset-sum has a solution $\iff$ 3SAT f is satisfiable
 - Take solution S to subset sum
 - For digit i where $1 \leq i \leq n$:
 - To get a 1 in digit i , include v_i or v_i' (not both)
 - If v_i in S then $x_i = T$
 - If v_i' in S then $x_i = F$
 - This provides an assignment

Proof: Satisfying Assignment

- Prove the assignment is satisfying
 - For digit $n+j$ where $1 \leq j \leq m$ to get a sum of 3 in digit $n+j$:
 - Need to include ≥ 1 literal of C_j and use S_j, S_j'
 - Therefore, C_j is satisfied

Reverse Implication

1. Subset-sum has a solution $\Leftrightarrow$ 3SAT f is satisfiable
 - Take satisfying assignment for f
 - If $x_i = T$: Add v_i to S
 - If $x_i = F$: Add v_i' to S
 - * i^{th} digit of t is correct
 - For clause C_j : ≥ 1 literal in C_j is satisfied
 - To get a sum of 3 in digit $n+j$, use S_j, S_j'

Knapsack is NP-complete

1. HW: Prove that knapsack (search) problem is NP-complete
 - Knapsack in NP
 - Known NP-complete problem $\rightarrow$ knapsack
 - Use subset-sum